COMPREHENSIVE EXAM PREPARATION

Grade 12 Geography Study Guide

A structured summary covering Units 1 through 5 for complete academic mastery.

UNIT 1: Major Geological Processes Associated with Plate Tectonics

The Earth is in constant motion, driven by internal (endogenic) and external (exogenic) forces. The Plate Tectonics Theory (late 1960s) explains how the rigid lithosphere slides on the ductile asthenosphere.

Continental Drift Theory

- **Proposed by:** Alfred Wegener (1912). Suggested all continents formed Pangaea, splitting into Laurasia and Gondwanaland.
- **Evidences:** Jigsaw fit of coastlines, matching rock structures/mountain chains, and fossil correlations (e.g., *Mesosaurus*).

Types of Plate Boundaries

- **Divergent (Constructive):** Plates move apart; creates oceanic ridges and rift valleys (e.g., East African Rift).
- **Convergent (Destructive):** Plates collide; denser plates subduct forming trenches, volcanic arcs, and fold mountains.
- **Transform (Conservative):** Plates slide horizontally past each other, forming fault zones and earthquakes.

> **Endogenic Forces Summary:** Folding creates anticlines and synclines (Young/Old Fold Mountains). Faulting creates Rift Valleys (Grabens) and Block Mountains (Horsts). Earthquakes release seismic energy (Body & Surface waves) measured via the Richter scale, occasionally triggering tsunamis.

Exogenic Forces & Weathering

- **Weathering:** Physical (thermal expansion, frost action, roots) and Chemical (solution, carbonation, hydrolysis forming caves and speleothems like Sof Omar).
- **Erosion & Mass Wasting:** Driven by running water, wind, glaciers, and gravity acting on unstable slopes.

UNIT 2: Climate Change

Climate change encompasses long-term shifts in temperature, precipitation, and wind systems driven by natural dynamics and intensified human activities.

Causes of Climate Change

- **Natural Drivers:** Milankovitch cycles (orbital eccentricity, axial tilt, precession), solar irradiance variations (sunspot cycles), and sulfur-rich volcanic aerosols causing cooling.
- **Anthropogenic Drivers:** Enhanced greenhouse effect from burning fossil fuels, deforestation, and industrial emissions trapping outgoing longwave radiation.

Consequences & Mitigation

- **Impacts:** Global surface temperature rise, melting cryosphere, accelerated sea-level rise, and ecological stress.
- **Mitigation vs. Adaptation:** Mitigation reduces emissions and builds carbon sinks; adaptation adjusts practices to cope with inevitable changes (e.g., drought-resistant crops).
- **Ethiopia's CRGE Strategy:** Focuses on agricultural efficiency, forest protection/re-expansion, renewable power generation, and green transport/industrial tech.

UNIT 3: Issues in Sustainable Development I (Resource Conflict Management)

Sustainable development meets present needs without compromising future generations by balancing economic, environmental, and social pillars.

Drivers of Resource Conflicts

- **Scarcity:** Competition over scarce renewable resources (demand-induced, supply-induced, and structural).
- **Governance:** Unclear ownership rights, inequitable benefit sharing, and lack of public participation.
- **Transboundary Dynamics:** Issues surrounding shared watercourses (e.g., Nile Basin riparian negotiations via the NBI).
- **Indigenous Resolution:** Cultural mechanisms offering win-win reconciliations (e.g., *Gadaa*, *Gumma*, *Mada'a*, *T'eer*, and *Shimagle*).

UNIT 4: Population Policies, Programs and the Environment

Population theories debate whether rapid growth hinders development or stimulates technological progress and economic innovation.

Population Theories

- **Malthusian & Neo-Malthusian:** Population grows geometrically while food grows arithmetically, causing poverty and degradation unless checked.
- **Boserup & Simon:** Optimistic view; human ingenuity and population pressure drive agricultural and technological innovation ("ultimate resource").
- **Marxist Perspective:** Attributes poverty and depletion to unequal capitalist resource distribution rather than population size.
- **Policies:** Pro-natalist policies incentivize population growth; anti-natalist policies curb fertility rates.

UNIT 5: Issues in Sustainable Development III (Challenges of Economic Development)

Economic development faces persistent barriers, including absolute/relative poverty, asymmetric globalization, trade imbalances, and systemic corruption.

Key Economic & Global Challenges

- **Poverty Definitions:** Absolute poverty is the inability to meet basic survival needs; relative poverty denotes income deprivation compared to societal medians.
- **Globalization Dynamics:** Increases international trade, capital mobility, and communication, but can also trigger labor exploitation, environmental stress, cultural erosion, and "brain drain".
- **Governance & Health:** Systemic corruption undermines public institutions and investment. Global health crises stem from ecological stress, population pressure, and weak sanitation infrastructure.

> **Study Tip for Exams:** Focus on comparing theoretical frameworks (e.g., Malthus vs. Boserup), understanding endogenic versus exogenic geological processes, and knowing Ethiopia's specific policy frameworks like the Climate-Resilient Green Economy (CRGE).